Aggregates

Notes from the lecture slides “Module 2 – Aggregates”

In this module

1. Why aggregates matter
2. Classification of aggregates
3. Mechanical properties
4. Physical properties
5. Deleterious substances
6. Sieve analysis and grading

1 Why aggregates matter

- Aggregate occupies about **three quarters of the volume** of concrete, so its properties strongly affect the concrete.
- It gives **dimensional stability**: aggregate does not shrink, so it restrains the shrinkage and creep of the cement paste.
- It is **cheaper** than cement paste, so more aggregate means cheaper concrete.
- It influences the **strength, durability and structural performance** of concrete.

1.1 How aggregate forms

- **Natural aggregate** comes from rock broken down by **weathering and abrasion** (river gravel and sand) or by crushing quarried rock.
- **Artificial aggregate** is manufactured, e.g. blast-furnace slag, expanded clay, recycled concrete.

Many properties are inherited from the **parent rock**: chemical and mineral composition, petrographic type, specific gravity, hardness, strength, physical and chemical stability, pore structure and colour.

Other properties depend on how the aggregate was produced: **particle shape and size, surface texture and absorption**.

2 Classification of aggregates

2.1 By size

Class	Size
Coarse aggregate	Larger than about 4–5 mm (IS 383 uses **4.75 mm**). Usual range 10–50 mm; **20 mm** is typical
Fine aggregate (sand)	Smaller than about 4–5 mm, down to 75 µm
Silt and clay	Finer than **75 µm** (clay is the finest fraction)

Common coarse aggregate fractions are 5–10 mm, 10–14 mm, 14–20 mm and above 20 mm. **5 mm (4.75 mm) is the dividing line** between fine and coarse aggregate.

2.2 Petrographic classification

Aggregates are grouped by the rock or mineral they consist of. ASTM C294 describes the important minerals:

- **Silica minerals**: quartz, opal, chalcedony, tridymite, cristobalite (opal, chalcedony, tridymite and cristobalite are the forms that can react with alkalis)
- Feldspars
- Micaceous minerals (mica)
- Carbonate minerals (calcite, dolomite)
- Sulphate minerals (gypsum)
- Iron sulphide minerals (pyrite)
- Ferromagnesian minerals
- Zeolites
- Iron oxide minerals
- Clay minerals

2.3 By shape (BS 812 Part 1)

The shape and texture of aggregate affect both fresh and hardened concrete. **Roundness** measures how sharp or angular the edges and corners of a particle are. For crushed aggregate, the shape depends on the parent rock, the type of crusher, and its **reduction ratio** (initial size ÷ size of crushed product).

Shape	Description	Examples
Rounded	Fully water-worn or completely shaped by attrition	River or seashore gravel; desert, seashore and wind-blown sand
Irregular	Naturally irregular, or partly shaped by attrition, with rounded edges	Other gravels; land or dug flint
Flaky	Thickness small compared with the other two dimensions	Laminated rock
Angular	Well-defined edges where roughly flat faces meet	Crushed rock of all types; talus; crushed slag
Elongated	Usually angular; length much greater than the other two dimensions ($l \gg b$, t)	–
Flaky and elongated	$l \gg b$ and $b \gg t$	–

The US classification describes degrees of wear: **well rounded** (no original faces left), **rounded** (faces almost gone), **subrounded** (considerable wear, faces reduced), **subangular** (some wear, faces untouched), **angular** (little evidence of wear).

2.4 Shape and packing

Size and shape decide how well particles pack together, and therefore how much cement paste is needed to fill the voids.

Angularity number

Angularity number = 67 − (% solid volume in a vessel filled with the aggregate)

- **67%** is the solid volume of the most rounded gravel, so the angularity number is the **percentage of voids in excess of that in rounded gravel**.

- The test uses closely sized fractions: 20–14 mm, 14–10 mm, 10–6.3 mm or 6.3–5 mm.
- It ranges from **0 (rounded) to about 11 (very angular)**. **Higher number = more angular**, and **workability falls as angularity number rises**.

Sphericity relates the **surface area of a particle to its volume**.

- It depends on the bedding and cleavage of the parent rock and on the crusher used.
- The **higher the surface-to-volume ratio, the lower the workability** (more surface to coat with paste).
- **Flaky particles harm durability**: they tend to lie in one plane, and water and air voids form underneath them.
- Flaky or elongated particles above about **10–15% of the mass** of coarse aggregate are undesirable.

Flakiness and elongation index

- **Flakiness index** = mass of flaky particles as a percentage of the sample mass. A particle is *flaky* if its least dimension is less than **0.6 × its mean sieve size**.
- **Elongation index** = mass of elongated particles as a percentage of the sample mass. A particle is *elongated* if its length is more than **1.8 × its mean sieve size**.
- IS 383:2016 limits the **combined** flakiness and elongation index to **40%**.

2.5 By surface texture (BS 812 Part 1)

Surface texture describes how polished or dull, smooth or rough the particle surface is. It depends on the rock's **hardness, grain size and pore structure** (hard, dense, fine-grained rocks break with smooth surfaces) and on the forces that have smoothed or roughened it.

Texture	Characteristics	Examples
Glassy	Conchoidal (shell-like) fracture	Black flint, vitreous slag
Smooth	Water-worn, or smooth because of fracture of laminated or fine-grained rock	Gravels, chert, slate, marble, some rhyolites
Granular	Fracture shows uniform, rounded grains	Sandstone, oolite
Rough	Rough fracture of fine- or medium-grained rock without visible crystals	Basalt, felsite, porphyry, limestone
Crystalline	Easily visible crystals	Granite, gabbro, gneiss
Honeycombed	Visible pores and cavities	Brick, pumice, foamed slag, clinker, expanded clay

> **Shape and texture vs water demand**
>
> Angular, rough, flaky particles pack loosely and have more surface area, so they need **more water** (or more paste) for the same workability. Rounded, smooth aggregate gives better workability; angular, rough aggregate gives better **bond**.

3 Mechanical properties

3.1 Bond

Bond is the adhesion between aggregate and cement paste.

- **Rougher texture** gives better mechanical interlock, so better bond.
- **Angular particles have more surface area**, which gives more bond.
- **Softer, porous and mineralogically heterogeneous** particles also bond better.
- **Check for good bond:** when a concrete specimen is crushed, some aggregate particles should be **broken right through**. If *too many* are broken, the aggregate is too weak.

3.2 Strength (crushing value)

- Crushing strength of rock used as aggregate is usually **80–200 MPa**, generally much higher than the concrete itself.
- A very **rigid** aggregate can cause cracking of the surrounding paste (because the paste and aggregate deform differently). Aggregate of **moderate strength and modulus** can help keep the concrete intact.

Aggregate crushing value test (IS 2386 Part 4):

1. Take aggregate **passing a 14 mm sieve and retained on 10 mm** (IS uses 12.5 mm to 10 mm). Larger sizes give higher crushing values.
2. **Oven-dry** at 100–110 °C for 4 hours.
3. Fill the **cylindrical mould** in three layers, tamping each layer, and place the **plunger** on top.
4. Load the assembly in a **compression testing machine** (to 400 kN over 10 minutes).
5. **Sieve on 2.36 mm.**

$$\text{Aggregate crushing value} = \frac{\text{mass passing 2.36 mm}}{\text{total mass of sample}} \times 100$$

A lower value means a stronger aggregate. IS 383 limits it to 45% for general concrete and 30% for wearing surfaces.

3.3 Toughness (impact value)

Toughness is the resistance of aggregate to **failure by impact**.

- A standard **hammer (132–137 N)** falls **15 times** under its own weight onto aggregate in a **cylindrical cup (102 mm dia., 50 mm deep)**.
- The fines passing 2.36 mm are weighed: aggregate impact value = mass passing 2.36 mm ÷ total mass × 100.

Maximum impact values:

- **25%** for heavy-duty concrete floor finishes
- **30%** for concrete pavement wearing surfaces

- **45%** for other concrete

3.4 Hardness (abrasion value)

Hardness is the resistance to **wear**. It matters for **roads, pavements and floors** carrying heavy traffic.

- Assessed by the **Los Angeles abrasion test** (IS 2386 Part 4:1963), which combines **attrition and abrasion**.
- Aggregate and an abrasive charge of **steel balls (about 47–48 mm dia., 390–445 g each; e.g. 11 balls for one grading)** are tumbled in a rotating steel **drum**. The wear is caused by the rubbing between the aggregate and the balls.
- After the specified revolutions, the sample is sieved (on 1.7 mm) and the loss is found:

$$\text{Abrasion value (\%)} = \frac{\text{original mass} - \text{mass retained}}{\text{original mass}} \times 100$$

Maximum permissible Los Angeles abrasion values for roads (IRC):

Type of pavement	Max. abrasion value (%)
Water bound macadam (WBM) sub-base course	60
WBM base course with bituminous surfacing	50
Bituminous bound macadam	50
WBM surfacing course	40
Bituminous penetration macadam	40
Bituminous surface dressing; cement concrete surface course	35
Bituminous concrete surface course	30

3.5 What controls aggregate properties

Aggregate properties fall into three groups depending on what controls them:

Controlled by	Properties
Porosity (microstructure)	Density, moisture absorption, strength, hardness, elastic modulus, soundness
Prior exposure and processing	Particle size, shape and texture
Chemical and mineral composition	Strength, hardness, elastic modulus, deleterious substances

These in turn affect the **fresh concrete** (consistency, cohesiveness, unit weight), the **mix proportions**, and the **hardened concrete** (ultimate strength, abrasion resistance, dimensional stability, durability).

4 Physical properties

4.1 Specific gravity

Specific gravity = mass of a given volume of material ÷ mass of the same volume of water.

Natural aggregates are **porous**, with both permeable and impermeable pores. Typical porosity:

- intrusive igneous rocks: about **2%**
- dense sedimentary rocks: about **5%**
- very porous sandstones and limestones: **10–40%**

So we define the volume in different ways:

- **Absolute specific gravity**: volume of solid material **excluding all pores**.
- **Apparent specific gravity**: volume of solid **including impermeable pores** but not the capillary (permeable) pores.
- **Bulk specific gravity (SSD)**: volume **including all pores**, with pores full of water. **This is what mix design needs**, because what matters is the space the particles occupy in concrete, pores included.

Typical apparent specific gravity: **2.6–2.7** for common rocks (2.65 for granite and sandstone, 2.60 for dense limestone).

4.2 Density and bulk density

- **Absolute density** refers to the volume of the individual particles; in practice particles can never be packed without voids.
- **Bulk density** is the **mass of aggregate that fills a unit volume** of a container. "Bulk" because the volume includes both the particles and the **voids between them**.
- Typical bulk density of normal-weight aggregate: **1300–1750 kg/m³**.

Void content from bulk density

$$\text{Voids (\%)} = \frac{G \times \rho_w - \text{bulk density}}{G \times \rho_w} \times 100$$

G = bulk specific gravity (SSD), ρ_w = 1000 kg/m³. The void ratio tells you **how much mortar is needed to fill the spaces between coarse aggregate particles**.

- Aggregate with **surface water packs less densely** (bulking), so the bulk density measured in the lab may not match site conditions.

4.3 Packing for maximum density

- Particles of **one size** can only pack so densely.
- **Smaller particles** can then fill the voids between the larger ones.
- **Maximum bulk density** is reached when fine aggregate is about **35–40% of the total aggregate mass**. In the slides' graph, voids fall from about 35% to a minimum of about 21–26% at around 40% sand, then rise again.

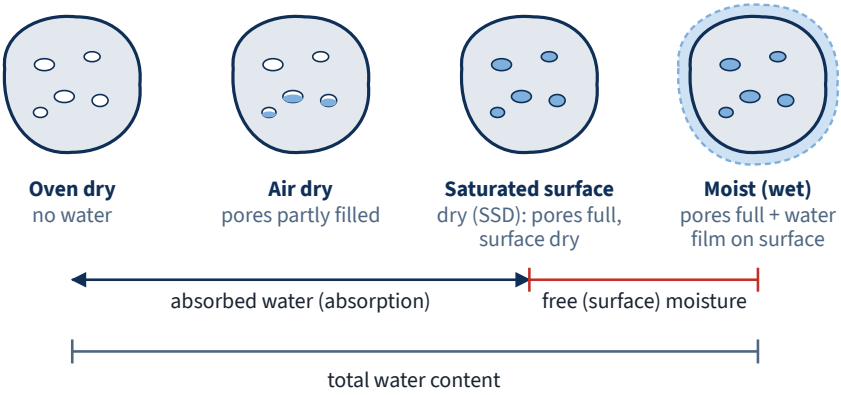
Maximum bulk density → densely packed structure → fewer voids → **less cement paste needed → more economical concrete**.

4.4 Porosity and absorption

The porosity, permeability and absorption of aggregate affect:

- the **bond** with cement paste,
- **resistance to freezing and thawing**,
- **chemical stability**,
- **resistance to abrasion**, and
- the **specific gravity**.

4.5 Moisture conditions



Moisture states of an aggregate particle.

- **Oven (bone) dry**: all water removed by prolonged drying.
- **Air dry**: some water has evaporated from the pores after standing in dry air; pores partly full.
- **Saturated surface dry (SSD)**: all pores full, but no water on the surface. This is the **reference state** for mix design, because SSD aggregate neither takes water from nor gives water to the mix.
- **Moist (wet)**: pores full **plus** a film of free water on the surface.

Absorption and moisture content

$$\text{Absorption (\%)} = \frac{W_{SSD} - W_{dry}}{W_{dry}} \times 100$$

Found by oven-drying an SSD sample for **24 hours** and measuring the loss of mass.

$$\text{Free (surface) moisture (\%)} = \frac{W_{moist} - W_{SSD}}{W_{SSD}} \times 100$$

Why moisture matters in batching

Dry aggregate **absorbs** some of the mixing water, so the mix becomes stiffer than designed. Wet aggregate **adds** its free moisture to the mix, raising the w/c ratio and lowering strength. So the mixing water and aggregate masses are **corrected** for the actual moisture state on site.

4.6 Bulking of sand

Moist sand occupies **more volume** than dry sand, because thin films of water around the grains push them apart. Bulking increases with moisture up to about **4–6%** moisture (volume can rise by 20–40%), then falls; **saturated sand does not bulk**. Finer sand bulks more. If sand is batched **by volume**, bulking must be allowed for, or the concrete will be short of sand.

4.7 Soundness and thermal properties

- **Unsound** aggregate changes volume excessively with wetting/drying or freezing/thawing, damaging the concrete. It is checked by the sodium or magnesium sulfate **soundness test** (IS 2386 Part 5).
- **Thermal properties** (expansion coefficient, conductivity) of aggregate largely control those of concrete; a large mismatch between aggregate and paste causes internal stress during temperature change.

5 Deleterious substances

5.1 Organic impurities

- Decayed vegetable matter (humus, organic loam), usually found **in sand rather than in coarse aggregate**. It interferes with hydration.
- **Colorimetric test**: the sand is shaken in a **3% NaOH solution** (which neutralizes the organic acids) and left for **24 hours**. The **darker the colour** of the liquid, the more organic matter.
- If in doubt, compare the **compressive strength** of concrete made with this aggregate against concrete made with aggregate of known quality.

5.2 Clay, silt and fine dust

- **Clay coatings** on aggregate surfaces **weaken the bond** with cement paste.
- **Silt and crusher dust** coatings have high fineness and surface area, which **increase water demand**.
- Measured by the **wet-sieving method** for coarse aggregate (ASTM C117) and the **decantation method** for fine aggregate (BS 812 Part 103).

5.3 Salt contamination

- Sand from the **seashore or river estuaries contains salt**; it can be removed by **washing in fresh water**.
- Salt (chlorides) is dangerous in **reinforced concrete** because it promotes **corrosion of steel**.
- Salt absorbs moisture from the air and causes **efflorescence**, unsightly white deposits on the concrete surface.

Deleterious substances affect both the fresh and the mechanical properties of concrete.

6 Sieve analysis and grading

6.1 Sieve analysis

Sieve analysis divides a sample of aggregate into fractions of the same particle size, to find its **grading** (particle size distribution). Grading decides the **void space between particles**, which affects the mechanical and durability properties of concrete.

The sample is shaken through a **stack of sieves** (coarsest at the top). The mass retained on each sieve is weighed and converted to percentages.

Sieve series	Coarse aggregate sieves	Fine aggregate sieves
BS (traditional)	75, 63, 50, 37.5, 28, 20, 14, 10, 6.3 mm	5, 2.36, 1.18 mm, 600, 300, 150 µm
ASTM	125, 100, 75, 63, 50, 37.5, 25, 19, 12.5, 9.5, 6.3 mm	4.75, 2.36, 1.18 mm, 600, 300, 150 µm (No. 4, 8, 16, 30, 50, 100)
BS EN	125, 63, 31.5, 16, 8 mm	4, 2, 1, 0.5, 0.25, 0.125, 0.063 mm
IS (IS 383)	80, 40, 20, 10, 4.75 mm	4.75, 2.36, 1.18 mm, 600, 300, 150 µm

6.2 Grading curve

- Grading is expressed as the **cumulative percentage passing** each sieve.
- The **grading curve** plots cumulative % passing (vertical) against **sieve size on a log scale** (horizontal).

6.3 Fineness modulus

The **fineness modulus (FM)** is an empirical index of how coarse or fine an aggregate is:

$$\text{FM} = \frac{\Sigma \text{ cumulative \% retained on the standard sieves}}{100}$$

Standard sieves: 150 µm, 300 µm, 600 µm, 1.18, 2.36, 4.75, 10, 20, 40, 80 mm (the pan is not included).

The higher the FM, the coarser the aggregate. Typical sand: fine about 2.2–2.6, medium 2.6–2.9, coarse 2.9–3.2.

Worked example (from the slides)

Sieve	Mass retained (g)	% retained	Cumulative % retained	% passing
10 mm	0	0.0	0	100
4.75 (5) mm	6	2.0	2	98
2.36 mm	31	10.1	12	88
1.18 mm	30	9.8	22	78
600 µm	59	19.2	41	59
300 µm	107	34.9	76	24
150 µm	53	17.3	93	7
Pan	21	6.8	–	–
Total	**307**		**246**	

FM = 246 / 100 = **2.46**, a fine-to-medium sand.

6.4 Why grading limits and maximum size are specified

Grading and maximum size affect the amount of **cement and water** needed, and through them the **workability, pumpability and durability**.

- **Larger aggregate has less surface area per unit volume** to be coated with paste, so it needs less cement and water.
- A **wide range of gradings** can be used without much effect on strength **for a given w/c ratio**.
- The aim is to **reduce the cement content without losing strength or workability**, which saves money.

Limits on maximum aggregate size:

- Not larger than **1/5 of the narrowest dimension of the formwork**.
- Not larger than **3/4 of the clear spacing between reinforcing bars**.

- **Larger particles produce more microcracks in the interfacial transition zone (ITZ)** between aggregate and paste, which is the weak link in concrete.
- In **high-strength concrete**, maximum size is usually limited to **19–20 mm**.

6.5 Types of grading

Grading	Description	Voids and permeability	Stability	Compaction
Dense (well) graded	Wide range of sizes; grain-to-grain contact	**Low**	High	Difficult
Open graded	Few fine particles; grain-to-grain contact	High	High	Difficult
Gap graded	Some **middle sizes missing**; no grain-to-grain contact	Moderate	Low	**Easy**
Uniform graded	Narrow range of sizes; grain-to-grain contact	High	Low	Difficult

Lowest void content comes from **smoothly graded coarse aggregate combined with a suitable proportion of graded sand**, which needs the least cement paste.

6.6 Gap-graded aggregate

Gap grading leaves out one or more intermediate sizes (the grading curve has a flat portion). It is used for:

- **preplaced aggregate concrete** (coarse aggregate placed first, then grouted);
- **exposed aggregate concrete**, for an attractive finish; and
- mixes of **relatively low workability compacted by vibration**, where it helps avoid segregation.

Quick revision

- Aggregate is ~75% of concrete volume: gives economy, dimensional stability, strength and durability.
- Coarse > 4.75 mm > fine > 75 µm > silt/clay. Shapes: rounded, irregular, flaky, angular, elongated.

- Angularity number = 67 − % solids (0 to 11; higher = more angular, less workable). Flaky if < 0.6 × mean size; elongated if > 1.8 × mean size.
- Crushing value and impact value: % passing 2.36 mm. Impact limits 25% (floors), 30% (pavements), 45% (others). LA abrasion: (W1 − W2)/W1 × 100.
- Specific gravity: absolute (no pores), apparent (impermeable pores), bulk SSD (all pores; used in mix design). Typical 2.6–2.7. Bulk density 1300–1750 kg/m^3.
- Moisture states: oven dry, air dry, SSD, moist. Absorption = (SSD − dry)/dry.
- Maximum packing at 35–40% sand. Bulking of moist sand up to 20–40%.
- Deleterious: organic matter (NaOH colour test), clay/silt coatings, salt (corrosion, efflorescence).
- FM = Σ cumulative % retained / 100; higher FM = coarser.
- Max size ≤ 1/5 narrowest form dimension, ≤ 3/4 clear bar spacing; 19 mm for HSC.
- Grading types: dense, open, gap (easy to compact, low stability), uniform.

Questions to practise

1. Why is aggregate used in concrete? How are aggregates classified by size and shape? (6 marks)
2. Define angularity number, flakiness index and elongation index. How do shape and texture affect concrete? (6 marks)
3. Describe the aggregate crushing value, impact value and Los Angeles abrasion tests. (10 marks)
4. Distinguish between absolute, apparent and bulk specific gravity. Which is used in mix design and why? (4 marks)
5. Explain the moisture states of aggregate with a sketch. Define absorption and free moisture. (5 marks)
6. What is bulking of sand? Why is it important? (3 marks)
7. List the deleterious substances in aggregate and their effects. How is organic matter detected? (5 marks)

8. A sieve analysis of 500 g of sand gave the following masses retained: 4.75 mm 10 g, 2.36 mm 40 g, 1.18 mm 75 g, 600 µm 120 g, 300 µm 150 g, 150 µm 80 g, pan 25 g. Find the fineness modulus. (5 marks)
9. What factors limit the maximum size of aggregate? (3 marks)
10. Compare dense, open, gap and uniform grading. Where is gap-graded aggregate used? (6 marks)